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October 27, 2025

Hiring Committee

Robotics Department – University of Michigan
Ann Arbor, MI

Dear Hiring Committee,

I'm writing to express my strong interest in the **Software Engineer position** within the Robotics Department at the University of Michigan. With over four years of experience building secure, data-driven applications across healthcare and financial domains, I'm deeply drawn to the University's mission of combining technological innovation with societal impact. The opportunity to support the Robotics Department's research and academic initiatives through reliable, user-focused software aligns perfectly with both my technical background and professional values.

In my current role at **Fidelity Investments**, I design and maintain cloud-native applications using **Java, Python, React, and AWS**, focusing on performance, maintainability, and compliance. Previously, as a **Full-Stack Developer at Lahey Hospital & Medical Center**, I developed HIPAA-compliant web applications and automated workflows that supported clinical and research teams. That experience honed my ability to translate non-technical requirements into scalable, secure solutions, a skill I believe is crucial when developing applications for students, faculty, and researchers in an academic environment.

I'm particularly drawn to this role's mix of **coding, application support, and collaboration**. I take pride in not only writing and testing code, but also ensuring that the people using the software can depend on it. My experience with tools like **Spring Boot, Node.js, SQL, MongoDB, and Docker** equips me to contribute to everything from database interactions to troubleshooting and documentation. I also bring a security-first mindset, having worked extensively with **OWASP** principles and institutional compliance frameworks—skills directly relevant to the University's commitment to secure and ethical software development.

I'm inspired by Michigan Engineering's vision to build a people-first future, and I'd be excited to contribute to that mission by creating reliable, inclusive, and maintainable tools that help your Robotics Department continue leading in research and innovation. I would welcome the opportunity to discuss how my background and passion for software craftsmanship can support your team's goals.

Thank you for considering my application.

Best regards,
Sai Kumar Bachireddy

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EDUCATION

University of Florida

Master of Science in Computer Science — GPA: 3.8

01/2023 – 12/2024

Gainesville, Florida, United States

SUMMARY

Full Stack Software Engineer with **4+ years** of experience building and deploying scalable enterprise applications across **Finance and Healthcare** domains. Proven expertise in **Java, Spring Boot, React, Angular, Node.js**, and **Cloud-native microservices** on **AWS and Azure**. Adept at optimizing application performance, streamlining **CI/CD pipelines**, and integrating **Machine-learning workflows** to enhance business outcomes. Recognized for reducing deployment cycles, improving system reliability, and delivering high-impact features in agile, cross-functional teams. **AWS Certified Solutions Architect** with a strong foundation in **MLOps, DevOps**, and **Distributed systems design**.

RELEVANT SKILLS

- **Languages & Frameworks:** Java, Python, J2EE, Spring Boot, Spring MVC, Hibernate, JavaScript, TypeScript, Node.js, RESTful APIs, Microservices, EJB, JSP, JSTL, Servlets, JAX-RS, JMS, JAXB
- **Frontend Technologies:** AngularJS, Angular (2+), ReactJS, Vue.js, HTML5, CSS3, Bootstrap, Redux, Backbone
- **Backend & API Development:** Node.js, RESTful & SOAP Services, Microservices Architecture, Monolithic Architecture, JDBC, JNDI, Custom Tag Libraries
- **Databases & Querying:** Oracle, MySQL, PostgreSQL, MongoDB, SQL Server, DB2, PL/SQL, Query Optimization, Database-Independent Persistence
- **Testing & Automation:** JUnit, Mockito, Selenium, Cypress, Jasmine, Karma, Postman, API Testing, TDD, BDD
- **DevOps & CI/CD:** Jenkins, GitLab, GitHub Actions, Docker, Kubernetes, Maven, Gradle, Ansible, Terraform
- **Cloud & MLOps:** AWS (EC2, S3, RDS, Lambda, API Gateway, CloudWatch, SNS, SQS), Azure, Google Cloud, ML Model Deployment, Versioning & Monitoring, Model Serving, Kubeflow, MLflow
- **Monitoring & Logging:** ELK Stack (Elasticsearch, Logstash, Kibana), Prometheus, Grafana, Log4j
- **API & Documentation Tools:** Swagger, Postman, Apigee, JSON, XML, YAML
- **Version Control & Tools:** Git, GitHub, Bitbucket, Jira, Confluence, Trello, IntelliJ IDEA, Eclipse, VSCode, Chrome DevTools, Build Deployment Tools

CERTIFICATIONS

- **AWS Certified Solutions Architect – Associate** 09/2024 – 09/2027
- **Apollo Certified GraphQL Developer – Associate** 05/2025

EXPERIENCE

Fidelity Investments

Java Full Stack Developer

01/2025 - present

Boston, Massachusetts

- Designed, developed, and maintained 25+ **Java APIs and microservices** Supporting Financial reporting / analytics using **Spring Boot 3.x, Spring MVC, Spring Core**, implementing Dependency Injection, **RESTful** services, and migrating legacy **SOAP services** to modern REST APIs reducing response time by 40%.
- Developed and deployed Full-stack web applications with **React, Redux, Node.js, jQuery, Knockout.js** components, integrated responsive UI, routing, and state management, serving 10M+ users.
- Developed and deployed **ML/AI pipelines** using **Airflow and MLflow**, incorporating **MLOps best practices** including continuous training, model versioning, monitoring, explainability, and production-scale deployment.
- Built and managed cloud-based infrastructure on **AWS** using Lambda, EC2, S3, RDS, SQS, API Gateway; containerized microservices supporting portfolio management and reporting workflows using Docker and Kubernetes, reducing deployment downtime by 35%.
- Designed and implemented scalable **Data integration pipelines** for cross-system synchronization using batch and streaming patterns (**Kafka, Airflow**), ensuring secure data exchange between internal services.
- Designed data models and schemas for efficient storage and retrieval in **NoSQL databases** like **Cassandra and MongoDB** handling 50M+ records, while ensuring database-independent persistence and scalability.

- Implemented **CI/CD pipelines** with **GitHub**, **GitLab CI/CD**, automated builds using **ANT**, **Maven**, and source code collaboration across distributed teams reducing deployment time by 30%.
- Monitored and resolved production and QA issues using **Jira**, **log analysis** using **log4j** and **ELK Stack**, automated notifications via **Apache Kafka**, and unit/integration testing using **JUnit**, **Mockito**, **Postman**, ensuring high-quality deliverables.
- Scripting and automation using **Bash**, **Python**, **PowerShell**, **Perl**, **Ruby**, supporting system administration, deployment automation, and operational efficiency for both traditional and ML-driven applications.

Lahey Hospital & Medical Center
Full Stack Developer

02/2024 – 12/2024
 Burlington, Massachusetts

- Developed and deployed 15+ **SOAP** and **RESTful** web services for **EHR (Epic)**-based patient management and **HL7** data exchange using JAX-WS, JAX-RS, and CXF; deployed on **Microsoft Azure** and Pivotal Cloud Foundry (PCF) to enable secure, **HIPAA**-compliant interoperability across **Healthcare systems**.
- Built and maintained microservices using Spring Boot and implemented database interactions with **SQL (Postgres)**, **JDBC**, and **NoSQL (MongoDB)** with DAO-based persistence ensuring horizontal scalability and fault tolerance and enabling faster data retrieval and ensuring high availability for critical healthcare operations.
- Established **SLO/SLI** metrics and implemented observability dashboards with **Grafana** and **Prometheus** to monitor system health and automate incident alerts, improving reliability and recovery time by 25%.
- Enhanced application security and compliance by performing **Threat modeling**, applying secure design principles, and enforcing role-based access controls and data encryption at rest and in transit.
- Designed responsive, user-friendly front-end interfaces using **Angular**, **HTML5**, **CSS3**, **Bootstrap**, integrating with backend microservices to deliver seamless customer and internal portal experiences for 50,000+ Users.
- Automated messaging and data workflows using **RabbitMQ (AMQP)** for asynchronous messaging, and developed **dynamic schemas** and **distributed data solutions** for high-performance applications.
- Enhanced application security by applying **OWASP** Top 10 principles and static code analysis (**SonarQube**, **Fortify**), enabling automated vulnerability gating and improving release security assurance.
- Worked in **Agile/SAFe methodology**, collaborated with clinical teams and IT analysts to gather requirements, provided technical guidance, and implemented version control, build automation using **Maven/ANT**, and logging via **Log4j** and **ELK Stack** improving deployment reliability and debugging efficiency by 40%.

BlackRock
Software Engineer

07/2021 – 11/2022
 Mumbai, India

- Built and maintained 10+ **Spring Boot** microservices exposing **REST** and **GraphQL APIs**, and deployed applications on **WebSphere** and **Linux environments**, for scalable, high-performance enterprise systems.
- Built **Front-end interfaces** using **React.js**, integrating reusable components from **NPM libraries** and implementing **Redux** for state management to deliver responsive and interactive web applications.
- Managed source code repositories on **GitHub**, and performed Continuous integration and Continuous deployment (CI/CD) using **AWS CodePipeline** and **Docker** for automated application deployment and configuration management reducing manual release effort by 40%.
- Configured and maintained **AWS services** and implemented **JMS server setup**, database connections, and handled database design with **MySQL**, including SQL queries optimization improving data throughput by 20%.
- Developed infrastructure automation scripts and playbooks using **Ansible with Python SSH**, reducing setup time by 30%, and monitored AWS metrics via CloudWatch to maintain 99.9% uptime.

Avanse Financial Services
Associate Software Developer

05/2020 – 06/2021
 Mumbai, India

- Developed and maintained **Enterprise applications** using J2EE technologies including Servlets, JSP, JDBC, EJB, XML, applying **Design patterns** such as **MVC**, **Singleton**, and implemented **Hibernate** for persistence with POJOs and DAOs for database operations.
- Built front-end interfaces using **AngularJS**, and **jQuery**, implementing controllers, services, and validations, and integrated **JasperReports** with RESTful services for on-demand report generation.
- Managed application deployment and cloud integration using **WebLogic Server** configured **JMS queues**, and monitored logs via **Linux servers (PuTTY)** for troubleshooting and performance tuning.
- Implemented **CI/CD pipelines** using **Jenkins**, and performed **Unit testing** and **Integration testing** with **JUnit** and **Mockito**, upping deployment frequency 30% and reducing QA effort 25%.
- Applied advanced Java features including **Collections**, **Multi-Threading**, **Exception Handling**, **Serialization**, and **Java 8** features (Lambda, Stream API, Method References) to optimize performance, enable bulk data processing, and enhance application scalability improving transaction processing speed by 35%.